

```
In [1]: import polars as pl
from Python import config
from Python.pipeline import training

pitcher_rolling = pl.read_parquet(config.PITCHER_ROLLING_PATH)
batter_rolling = pl.read_parquet(config.BATTER_ROLLING_PATH)
park_factors = (
    pl.read_parquet(config.PARK_FACTORS_PATH)
    if config.PARK_FACTORS_PATH.exists()
    else None
)
pitcher_rolling.shape, batter_rolling.shape, (park_factors.shape if park_factors is
```

```
Out[1]: ((14124, 236), (146385, 20), (120, 5))
```

```
In [2]: print(training._LINEUP_RATE_COLUMNS)

{'opp_lineup_whiff': 'whiff_rate_std', 'opp_lineup_chase': 'chase_rate_std'}
```

```
In [3]: lineup = training.opposing_lineup_features(pitcher_rolling, batter_rolling)
lineup.shape
lineup.head()
```

```
Out[3]: shape: (5, 6)
```

game_pk	pitcher	opp_lineup_k	opp_lineup_k_vs_hand	opp_lineup_whiff	opp_lineup_chase
i64	i64	f64	f64	f64	f64
745837	500779	0.24759	0.323145	0.125404	0.26515
718524	519242	0.239673	0.26002	0.150369	0.302363
746190	519242	0.218462	0.193639	0.110372	0.280802
746748	543135	0.23481	0.240552	0.125228	0.308607
718317	571578	0.209329	0.236299	0.111429	0.247134

```
In [4]: check = pl.DataFrame({
    "pitcher_rolling_rows": [pitcher_rolling.height],
    "lineup_rows": [lineup.height],
    "unique_game_pitcher_pairs": [pitcher_rolling.select("game_pk", "pitcher").unique().height]
})
check
```

```
Out[4]: shape: (1, 3)
```

pitcher_rolling_rows	lineup_rows	unique_game_pitcher_pairs
i64	i64	i64
14124	14124	14124

```
In [5]: lineup.select(pl.all().exclude("game_pk", "pitcher").null_count())
```

Out[5]: shape: (1, 4)

opp_lineup_k	opp_lineup_k_vs_hand	opp_lineup_whiff	opp_lineup_chase
u32	u32	u32	u32
90	99	90	90

```
In [6]: pitcher_training = training.build_pitcher_training(pitcher_rolling, batter_rolling,  
pitcher_training.shape  
pitcher_training.columns
```

```
Out[6]: ['game_pk',
         'pitcher',
         'game_date',
         'season',
         'player_name',
         'home_team',
         'away_team',
         'p_throws',
         'PA',
         'K',
         'Outs',
         'is_home',
         'opp_team',
         'k_rate',
         'pitcher_name',
         'k_rate_P5',
         'k_rate_P10',
         'k_rate_P20',
         'k_rate_std',
         'bb_rate_P5',
         'bb_rate_P10',
         'bb_rate_P20',
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'xBA_P5',
'xBA_P10',

```

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'xWOBA_P5',
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'FIP_P5',
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'xFIP_P3',
'xFIP_P5',
'xFIP_P10',
'opp_lineup_k',
'opp_lineup_k_vs_hand',
'opp_lineup_whiff',
'opp_lineup_chase',
'park_k_factor']

```

```

In [7]: pitcher_training_no_park = training.build_pitcher_training(pitcher_rolling, batter_
"park_k_factor" in pitcher_training_no_park.columns

```

Out[7]: False

```

In [8]: lineup_source_cols = {"k_rate_std", "k_rate_std_vL", "k_rate_std_vR", *training._LI
non_rolling_leak_risk = [c for c in lineup_source_cols if not c.endswith(("_std", "
non_rolling_leak_risk

```

Out[8]: []

```

In [9]: if park_factors is not None:
    spot = (
        pitcher_training
        .filter(pl.col("season") == 2025)
        .select("game_pk", "pitcher", "home_team", "season", "park_k_factor")
        .head(5)
    )
    park_factors.filter(pl.col("season") == 2025).select("home_team", "park_k_facto

```

```

In [10]: paths = training.run()
paths

```

[level 3] wrote pitcher_training: C:\Users\ckaplinger\Downloads\Personal-Projects\MLB-Props\Data\processed\pitcher_training.parquet

[level 3] wrote batter_training: C:\Users\ckaplinger\Downloads\Personal-Projects\MLB-Props\Data\processed\batter_training.parquet

```

Out[10]: {'pitcher_training': WindowsPath('C:/Users/ckaplinger/Downloads/Personal-Projects/MLB-Props/Data/processed/pitcher_training.parquet'),
'batter_training': WindowsPath('C:/Users/ckaplinger/Downloads/Personal-Projects/MLB-Props/Data/processed/batter_training.parquet')}

```

```

In [11]: pitcher_training_final = pl.read_parquet(config.PITCHER_TRAINING_PATH)
batter_training_final = pl.read_parquet(config.BATTER_TRAINING_PATH)

label_cols = ["K", "PA", "Outs", "k_rate"]
[c for c in label_cols if c in pitcher_training_final.columns]

```

```
Out[11]: ['K', 'PA', 'Outs', 'k_rate']
```